Interchangeable-Insert Water Cold-Plate Testbed

Rev A Preliminary Engineering Validation Report

NOT MACHINING APPROVED

Scope

This report records deterministic dimensional screening, CAD checks, export/reimport checks, inspection intent, departures and unresolved physical release work. The model is regenerable from one shared parameter file used by both CAD and calculations.

Datum / axis	CP-001 upper sealing deck z=0; cavity center x=y=0; ports on x-axis
Load case	30 kPa coolant gauge pressure (analysis only)
Ordinary tolerance	+/-0.10 mm unless overridden
Temperature range	20-60 C component envelope
Selected assembly	FI-100; all four inserts exported individually

External Seal

Check	Result
nominal squeeze percent	19.08
minimum squeeze with 0p03 opening percent	17.18
maximum squeeze nominal cs percent	19.85
nominal 2d gland fill percent	68.73
groove centerline diameter	172.7
oring centerline diameter	173.7
centerline fit percent	-0.6274
status	PRELIMINARY: verify actual ring tolerance, compound, swelling, thermal exposure, and Parker axial-face-seal rules

Thermal Insert Fit

Check	Result
cold minimum diametral clearance	1.35
hot insert cold base clearance	0.499
clearance after reserved growth	0.299
hot insert max diameter	164.5
status	passes dimensional screening; conditioned-polymer measurement remains required

Assembled Height Budget

Check	Result
nominal channel height	4
worst case min	3.95
worst case max	4.05
specified range	3.95 to 4.05
status	exactly consumes tolerance band before flatness; match-finish insert floor to measured base

Pressure Screening

Check	Result
pressure mpa	0.03
resultant force n	641.5
load per fastener from pressure only n	40.09
ideal clamped deflection mm	0.02072
simply supported screen mm	0.08289
allowable additional opening mm	0.1
status	screen only; gasket reaction, preload, flange compliance, contact, and thermal distortion unresolved

Hardware Stack

Check	Result
nominal engagement mm	10.5
worst case min engagement mm	10.25
worst case max engagement mm	10.75
modeled full thread depth mm	12
drill point clearance to port envelope mm	1
status	select actual M5x25 screw/washer and verify thread runout; torque not specified

Port Screening

Check	Result
thread envelope bottom wall mm	4.15
drilled passage bottom wall mm	8
candidate thread length mm	13.5
status	actual 1/4-18 NPT fitting geometry and engagement required before release

Internal Gasket

Check	Result
nominal compression percent	25
service compression range percent	12.5 to 32.5
status	geometry only; compound, force/deflection, compression set, bulging, and retention test unresolved

Flow

Check	Result
fi100 cad centerline path length mm	1120
fi100 note	9 x 96 straight passes, 8 semicircular R6 returns, and two modeled diagonal feed transitions
fi200 parallel channel length mm	96
fi200 channel count	8
fi300 status	tapered-manifold candidate; hydraulic improvement not demonstrated

STEP Export/Reimport Validation

Every individual STEP and each assembly STEP was reimported. Individual files were checked against source bounding dimensions and volume. Assembly files were checked for a minimum expected solid count plus aggregate envelope and volume.

File	Src	Reimport	Volume mm3	Result
CP-001_base.step	1	1	879399.28	PASS
CP-002_cover.step	1	1	217909.27	PASS
CL-002_structural_clamp.step	1	1	143819.48	PASS
TH-001_payload.step	1	1	40212.39	PASS
CL-001_payload_clamp.step	1	1	13096.63	PASS
BR-001_bench_bracket.step	1	1	21720.60	PASS
GS-001_AS568-166_envelope.step	1	1	2942.67	PASS
GS-013_transfer_orings_2x.step	2	2	197.01	PASS
HW-M5x25_screw_envelope.step	1	1	774.60	PASS
HW-M5_washer_envelope.step	1	1	56.48	PASS
HW-NPT-1_4_candidate_envelope.step	1	1	7065.72	PASS
HT-001_heater_provisional_envelope.step	1	1	17671.46	PASS
FI-000_insert.step	1	1	25802.31	PASS
FI-000_rib_gasket.step	1	1	1470.27	PASS
FI-000_fluid_domain.step	1	1	72583.36	PASS
FI-100_insert.step	1	1	48791.61	PASS
FI-100_rib_gasket.step	3	3	4340.13	PASS
FI-100_fluid_domain.step	1	1	35647.30	PASS
FI-200_insert.step	1	1	49425.51	PASS
FI-200_rib_gasket.step	8	8	3486.27	PASS
FI-200_fluid_domain.step	1	1	34633.06	PASS
FI-300_insert.step	1	1	50855.56	PASS
FI-300_rib_gasket.step	8	8	3486.27	PASS
FI-300_fluid_domain.step	1	1	32344.98	PASS
CP-REV-A_assembly_FI-100.step	-	51	1482690.98	PASS
CP-REV-A_exploded_FI-100.step	-	13	1368379.93	PASS
CP-REV-A_insert_family.step	-	24	187657.90	PASS

Fluid connectivity

FI-000: 1 connected fluid solid - PASS

FI-100: 1 connected fluid solid - PASS

FI-200: 1 connected fluid solid - PASS

FI-300: 1 connected fluid solid - PASS

Prototype mesh and gasket-outline validation

FI-000_prototype.stl: 2372 triangles, 0 open/non-manifold edges - PASS

FI-100_prototype.stl: 6756 triangles, 0 open/non-manifold edges - PASS

FI-200_prototype.stl: 2896 triangles, 0 open/non-manifold edges - PASS

FI-300_prototype.stl: 2944 triangles, 0 open/non-manifold edges - PASS

FI-000_gasket_cutting_outline.dxf: 1 imported outline entities - PASS

FI-100_gasket_cutting_outline.dxf: 3 imported outline entities - PASS

FI-200_gasket_cutting_outline.dxf: 1 imported outline entities - PASS

FI-300_gasket_cutting_outline.dxf: 1 imported outline entities - PASS

Specification Departures and Interpretations

Item	Departure / interpretation	Engineering reason
References	No separate review or original plan was attached; this package uses the pasted Rev A brief as the controlling input.	Avoid inventing absent source requirements.
FI-100 feeds	Two diagonal feed transitions join the +/-70 transfer centers to the first/last 96 mm pass.	Maintains a continuous path inside the usable disk without crossing intermediate passages.
FI-100 length	CAD centerline screening length 1120.4 mm, not 1.5 m.	Measured from the modeled routing definition.
Static taps	Two 1.5 mm bores with 5 mm thread envelopes run from the front side to the transfer centers.	Defines flush upstream/downstream measurement locations without penetrating CP-002.
Locator	Round pin at y=+70 and radial slot/pin at y=-70 instead of a central pin.	Avoids the active central flow/ROI while providing one constrained and one expansion-permitting datum.
CL-001	Envelope-level ring with three adjustment slots only.	Actual heater thickness, lead exit and mounting remain unresolved; no false detail added.
NPT fittings	Conservative cylindrical/hex envelope, not supplier-specific thread geometry.	Actual fitting was not selected.
Gasket corners	CAD cutting outlines are nominal planar geometry without die-cut corner compensation.	Supplier/process-specific compensation is not yet available.

Physical Release Checks Still Required

1. Select and lot-verify main and transfer O-rings; recalculate Parker axial-gland squeeze, circumferential fit, volume fill, swelling and thermal effects using supplied tolerances.

2. Select rib-gasket compound and retention process; test force/deflection, lateral bulging, movement, compression set and durability at temperature and coolant exposure.

3. Run contact/nonlinear structural analysis or a load test including gasket reaction, fastener preload, cover/clamp/base compliance and thermal distortion. Confirm additional opening <=0.10 mm and local main-seal opening <=0.03 mm.

4. Condition representative POM-C material and measure OD, locator displacement, port registration and gasket registration over 20-60 C.

5. Select heater, define resistive pattern, thickness, lead exit, electrical isolation, mounting, coating, sensor locations, clamp contacts and thermal interface resistance.

6. Select actual M5 hardware, washer and NPT fittings; verify full thread depth, drill-point clearance, engagement, passage intersections, torque/preload method and tool access.

7. Pressure/leak test the assembled loop; determine maximum achievable loop pressure and explicitly establish the operating rating separately from the 30 kPa analysis case.

8. Demonstrate thermal observability and camera access with the real heater, coating, sensors, hoses, brackets and optical setup.

Conclusion

The package is suitable as a regenerable preliminary engineering assembly and prototype basis. It is explicitly not machining-approved or pressure-rated. The unresolved seal, structural clamp and heater-interface items above are release blockers, not optional refinements.